

Titanic Survival Analysis

Task 4: Data Storytelling And Statistical Validation

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APEX PLANET

Business Objective

- ▶ The objective of this analysis is to understand
- ▶ the factors that influenced passenger survival
- ▶ on the Titanic and identify key patterns using
- ▶ data analysis and statistical validation.



Dataset Overview

- ▶ **Dataset: Titanic Dataset**
- ▶ **Key Features:**
 - ▶ • **Passenger Class**
 - ▶ • **Gender**
 - ▶ • **Age**
 - ▶ • **Fare**
 - ▶ • **Survival Status**
- ▶ **Total Records: 891**

Key Insights

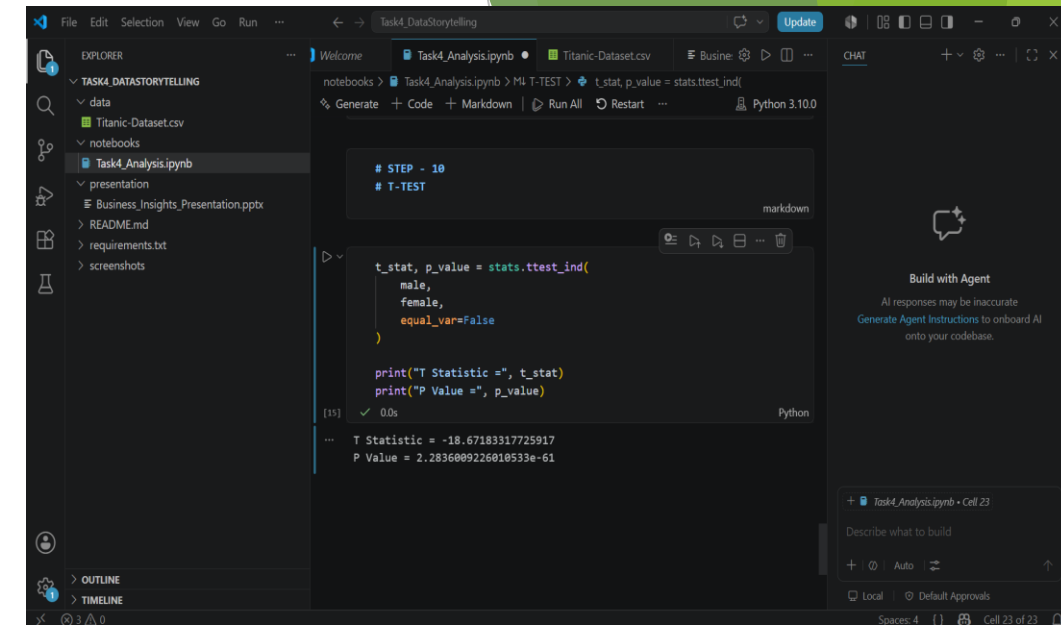
- ▶ • **Female passengers had higher survival rates.**
- ▶ • **First-class passengers survived more often**
than second and third-class passengers.
- ▶ • **Passenger class influenced survival chances.**
- ▶ • **Survival varied significantly across groups.**

Data Story

- ▶ **Analysis revealed that survival was strongly associated with gender and passenger class.**
- ▶ **Female passengers had a much higher chance of survival compared to males.**
- ▶ **Passengers travelling in first class also experienced better survival outcomes.**
- ▶ **These findings suggest that social and economic factors played an important role during the disaster.**

Statistical Validation

- ▶ Hypothesis:
- ▶ H_0 : Male and female survival rates are equal.
- ▶ H_1 : Male and female survival rates are different.
- ▶ T-Test was performed to validate the hypothesis.



This screenshot shows a Jupyter Notebook interface with a file explorer on the left containing folders for 'TASK4_DATASTORYTELLING', 'data', 'notebooks', 'presentation', and 'screenshots'. The main editor displays a Python cell for a T-test. The code defines a function `t_stat, p_value = stats.ttest_ind(male, female, equal_var=False)` and prints the results. The output shows a T Statistic of -18.67183317725917 and a P Value of 2.2836009226010533e-61. A right-hand sidebar contains a 'CHAT' section with a 'Build with Agent' button and a 'Generate Agent Instructions' link.

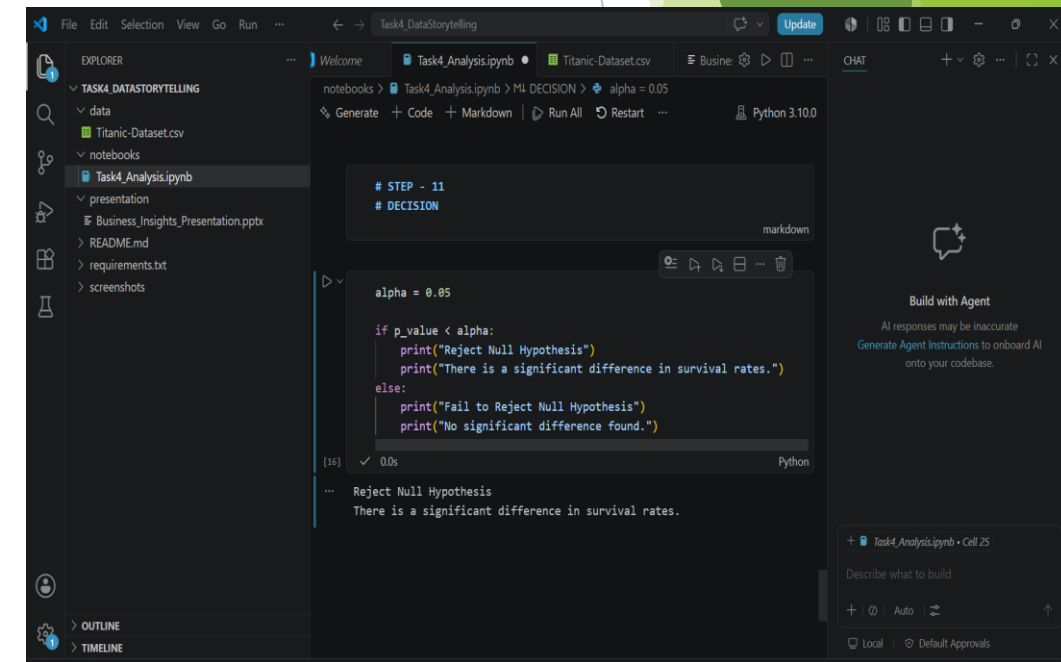
```
# STEP - 10
# T-TEST

t_stat, p_value = stats.ttest_ind(
    male,
    female,
    equal_var=False
)

print("T Statistic =", t_stat)
print("P Value =", p_value)
```

[15] ✓ 0.0s Python

T Statistic = -18.67183317725917
P Value = 2.2836009226010533e-61



This screenshot shows the next step in the Jupyter Notebook. The code sets `alpha = 0.05` and uses an if-else statement to print a decision based on the p-value. The output indicates that the null hypothesis is rejected because the p-value is less than the alpha level. The right-hand sidebar is identical to the one in the previous screenshot.

```
# STEP - 11
# DECISION

alpha = 0.05

if p_value < alpha:
    print("Reject Null Hypothesis")
    print("There is a significant difference in survival rates.")
else:
    print("Fail to Reject Null Hypothesis")
    print("No significant difference found.")
```

[16] ✓ 0.0s Python

Reject Null Hypothesis
There is a significant difference in survival rates.

Recommendations & Conclusion

Recommendations

- ▶ • Improve emergency preparedness.
- ▶ • Ensure equal access to safety measures.
- ▶ • Use data-driven planning for passenger safety.

Conclusion

- ▶ The analysis showed that gender and passenger class significantly influenced survival.
- ▶ Statistical testing confirmed these findings, demonstrating the value of data storytelling
and hypothesis testing.